

# UW Quantum REU Solution

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**Problem 1** (Distinguishing between two states). Let  $n \in \mathbb{N}$ , Suppose you are handed *one* of two  $n$ -qubit states  $|\psi_0\rangle, |\psi_1\rangle$  uniformly at random. Assume that you know exactly what the two states are: you just don't know *which one* of the two you are given. Assume also that the two states are not identical (differing only by a global phase still counts as being identical here). In this question you will find the projective measurement  $\{P, \mathbb{I} - P\}$  that maximizes the distinguishing success probability, i.e. it maximizes

$$\frac{1}{2}\Pr[0 | \psi_0] + \frac{1}{2}\Pr[1 | \psi_1] \quad (0.1)$$

where  $\Pr[i | \psi_i]$  denotes the probability that the measurement outcome is  $i$ , given that the state was  $|\psi_i\rangle$ .

- (i) *Carefully* show that, without loss of generality, you can restrict your search to the 2-dimensional space spanned by  $|\psi_0\rangle$  and  $|\psi_1\rangle$ , i.e. you can restrict yourself to consider measurements of the form

$$\{|u\rangle\langle u|, \mathbb{I} - |u\rangle\langle u|\} \quad (0.2)$$

where  $|u\rangle$  lies in the span of  $|\psi_0\rangle$  and  $|\psi_1\rangle$

(You should try to make this proof as rigorous as possible – this question is more subtle than meets the eye!).

- (ii) Find the optimal measurement, and show that it is optimal.

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*Solution.*

Let  $\mathcal{H}$  denote the Hilbert Space of the  $n$ -qubit system (which  $\dim \mathcal{H} = 2^n$ ). Let  $V = \text{span}\{|\psi_0\rangle, |\psi_1\rangle\}$  be the 2-dimensional linear subspace of  $\mathcal{H}$  (by the assumption that  $|\psi_0\rangle, |\psi_1\rangle$  are distinct states, hence they're linearly independent), then  $\mathcal{H} = V \oplus V^\perp$ .

Here, if  $|\varphi\rangle$  is a normalized state, we'll use  $\langle\varphi|u\rangle\langle u|\varphi\rangle =: \Pr[1 | \varphi]$  (probability of having measurement outcome 1 given state  $|\varphi\rangle$ ), while  $\langle\varphi|(\mathbb{I} - |u\rangle\langle u|)|\varphi\rangle = 1 - \langle\varphi|u\rangle\langle u|\varphi\rangle =: \Pr[0 | \varphi]$  (probability of having measurement outcome 0 given state  $|\varphi\rangle$ ).

- (i) Let  $|u\rangle \in \mathcal{H}$  be any normalized states, then notice that  $|u\rangle = |v\rangle + |w\rangle$  for unique  $|v\rangle \in V$  and  $|w\rangle \in V^\perp$ . Which, since  $|\psi_0\rangle, |\psi_1\rangle, |v\rangle \in V$  while  $|w\rangle \in V^\perp$ , one has  $\langle w|\psi_0\rangle = \langle w|\psi_1\rangle = \langle w|v\rangle = 0$ . Also, the normalization condition provides the following equality:

$$1 = \langle u|u\rangle = \langle v|v\rangle + \langle w|w\rangle + \langle v|w\rangle = \langle v|v\rangle + \langle w|w\rangle \quad (0.3)$$

Hence, the inner product property guarantees  $0 \leq \langle v|v\rangle \leq 1$ , and  $\langle v|v\rangle = 1$  iff  $\langle w|w\rangle = 0$  iff  $|w\rangle = 0$  iff  $|u\rangle = |v\rangle \in V$ .

Now, write the corresponding projection operator, one has the following:

$$|u\rangle\langle u| = |v\rangle\langle v| + |v\rangle\langle w| + |w\rangle\langle v| + |w\rangle\langle w| \quad (0.4)$$

Hence, the component of the distinguishing success probability –  $\Pr[0 \mid \psi_0]$  and  $\Pr[1 \mid \psi_1]$  are given as follow:

$$\Pr[1 \mid \psi_1] = \langle \psi_1 | u \rangle \langle u | \psi_1 \rangle = \langle \psi_1 | v \rangle \langle v | \psi_1 \rangle \quad (0.5)$$

$$\Pr[0 \mid \psi_0] = 1 - \langle \psi_0 | u \rangle \langle u | \psi_0 \rangle = 1 - \langle \psi_0 | v \rangle \langle v | \psi_0 \rangle \quad (0.6)$$

Which, this is the consequence of  $\langle w | \psi_0 \rangle = \langle w | \psi_1 \rangle = 0$ .

Then, the distinguishing success probability becomes:

$$\left( \frac{1}{2} \Pr[0 \mid \psi_0] + \frac{1}{2} \Pr[1 \mid \psi_1] \right)_{|u\rangle} = \frac{1}{2} + \frac{1}{2} (\langle \psi_1 | v \rangle \langle v | \psi_1 \rangle - \langle \psi_0 | v \rangle \langle v | \psi_0 \rangle) \quad (0.7)$$

Now, there are two cases to consider:

- If  $|v\rangle = 0$ , then the above probability simply reduces to  $\frac{1}{2}$ . Which, let's consider  $|u'\rangle = |\psi_0\rangle + |\psi_1\rangle \in V$  (which is a nonzero vector by linear independence of  $|\psi_0\rangle, |\psi_1\rangle$ ), and consider  $|v'\rangle := \frac{|u'\rangle}{\sqrt{\langle u' | u' \rangle}}$  (the normalized version of  $|u'\rangle$ ). Notice with respect to this state, one has the distinguishing success probability as follow:

$$\begin{aligned} \left( \frac{1}{2} \Pr[0 \mid \psi_0] + \frac{1}{2} \Pr[1 \mid \psi_1] \right)_{|v'\rangle} &= \frac{1}{2} + \frac{1}{2} (\langle \psi_1 | v' \rangle \langle v' | \psi_1 \rangle - \langle \psi_0 | v' \rangle \langle v' | \psi_0 \rangle) \\ &= \frac{1}{2} + \frac{1}{2 \langle u' | u' \rangle} (\langle \psi_1 | u' \rangle \langle u' | \psi_1 \rangle - \langle \psi_0 | u' \rangle \langle u' | \psi_0 \rangle) \\ &= \frac{1}{2} + \frac{1}{2 \langle u' | u' \rangle} (\langle \psi_1 | (|\psi_0\rangle \langle \psi_0| + |\psi_0\rangle \langle \psi_1| + |\psi_1\rangle \langle \psi_0| + |\psi_1\rangle \langle \psi_1|) | \psi_1 \rangle) \\ &\quad - \frac{1}{2 \langle u' | u' \rangle} (\langle \psi_0 | (|\psi_0\rangle \langle \psi_0| + |\psi_0\rangle \langle \psi_1| + |\psi_1\rangle \langle \psi_0| + |\psi_1\rangle \langle \psi_1|) | \psi_0 \rangle) \\ &= \frac{1}{2} + \frac{1}{2 \langle u' | u' \rangle} (|\langle \psi_0 | \psi_1 \rangle|^2 + \langle \psi_1 | \psi_0 \rangle + \langle \psi_0 | \psi_1 \rangle + 1) \\ &\quad - \frac{1}{2 \langle u' | u' \rangle} (1 + \langle \psi_1 | \psi_0 \rangle + \langle \psi_0 | \psi_1 \rangle + |\langle \psi_0 | \psi_1 \rangle|^2) \\ &= \frac{1}{2} \end{aligned}$$

Hence, considering the normalized state  $|v'\rangle \in V$ , one can achieve the same distinguishing success probability.

- If  $|v\rangle \neq 0$ , consider the new state  $|u'\rangle := \frac{|v\rangle}{\sqrt{\langle v | v \rangle}} \in V$  (i.e. the normalized  $|v\rangle$ ), and also any normalized vector  $|w'\rangle \in V$  such that  $\langle u' | w' \rangle = 0$  (i.e. orthogonal to  $|u'\rangle$ ).

Which,  $\{|u'\rangle, |w'\rangle\} \subset V$  is an orthonormal list, hence linearly independent; also, with  $\dim V = 2$ , they form a basis. Hence, for any vector  $|\varphi\rangle \in V$ , its decomposition becomes  $|\varphi\rangle = \langle u' | \varphi \rangle |u'\rangle + \langle w' | \varphi \rangle |w'\rangle$ , implying that  $\langle \varphi | \varphi \rangle = |\langle u' | \varphi \rangle|^2 + |\langle w' | \varphi \rangle|^2$  by the orthonormality, or  $|\langle w' | \varphi \rangle|^2 = \langle \varphi | \varphi \rangle - |\langle u' | \varphi \rangle|^2$ . In particular if choosing  $|\varphi\rangle \in V$  as a normalized vector (for instance,  $|\varphi\rangle = |\psi_0\rangle, |\psi_1\rangle$ ), this reduces to  $|\langle w' | \varphi \rangle|^2 = 1 - |\langle u' | \varphi \rangle|^2$ .

Then, notice the new distinguishing successive probability of  $|u'\rangle, |w'\rangle$  becomes the following:

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$$\begin{aligned}
\left(\frac{1}{2}\Pr[0 \mid \psi_0] + \frac{1}{2}\Pr[1 \mid \psi_1]\right)_{|u'\rangle} &= \frac{1}{2} + \frac{1}{2}(\langle\psi_1|u'\rangle\langle u'|\psi_1\rangle - \langle\psi_0|u'\rangle\langle u'|\psi_0\rangle) \\
&= \frac{1}{2} + \frac{1}{2\langle v|v\rangle}(\langle\psi_1|v\rangle\langle v|\psi_1\rangle - \langle\psi_0|v\rangle\langle v|\psi_0\rangle) \tag{0.9}
\end{aligned}$$

$$\begin{aligned}
\left(\frac{1}{2}\Pr[0 \mid \psi_0] + \frac{1}{2}\Pr[1 \mid \psi_1]\right)_{|w'\rangle} &= \frac{1}{2} + \frac{1}{2}(\langle\psi_1|w'\rangle\langle w'|\psi_1\rangle - \langle\psi_0|w'\rangle\langle w'|\psi_0\rangle) \\
&= \frac{1}{2} + \frac{1}{2}(|\langle w'|\psi_1\rangle|^2 - |\langle w'|\psi_0\rangle|^2) \\
&= \frac{1}{2} + \frac{1}{2}((1 - |\langle u'|\psi_1\rangle|^2) - (1 - |\langle u'|\psi_0\rangle|^2)) \tag{0.10} \\
&= \frac{1}{2} - \frac{1}{2}(\langle\psi_1|u'\rangle\langle u'|\psi_1\rangle - \langle\psi_0|u'\rangle\langle u'|\psi_0\rangle) \\
&= \frac{1}{2} - \frac{1}{2\langle v|v\rangle}(\langle\psi_1|v\rangle\langle v|\psi_1\rangle - \langle\psi_0|v\rangle\langle v|\psi_0\rangle)
\end{aligned}$$

Which, if  $(\langle\psi_1|v\rangle\langle v|\psi_1\rangle - \langle\psi_0|v\rangle\langle v|\psi_0\rangle) \geq 0$ , then the following inequality holds because  $\langle v|v\rangle \leq 1$ :

$$\begin{aligned}
\left(\frac{1}{2}\Pr[0 \mid \psi_0] + \frac{1}{2}\Pr[1 \mid \psi_1]\right)_{|u'\rangle} &= \frac{1}{2} + \frac{1}{2\langle v|v\rangle}(\langle\psi_1|v\rangle\langle v|\psi_1\rangle - \langle\psi_0|v\rangle\langle v|\psi_0\rangle) \\
&\geq \frac{1}{2} + \frac{1}{2}(\langle\psi_1|v\rangle\langle v|\psi_1\rangle - \langle\psi_0|v\rangle\langle v|\psi_0\rangle) \tag{0.11} \\
&= \left(\frac{1}{2}\Pr[0 \mid \psi_0] + \frac{1}{2}\Pr[1 \mid \psi_1]\right)_{|u\rangle}
\end{aligned}$$

Else if  $(\langle\psi_1|v\rangle\langle v|\psi_1\rangle - \langle\psi_0|v\rangle\langle v|\psi_0\rangle) < 0$ , then the following inequality holds:

$$\begin{aligned}
\left(\frac{1}{2}\Pr[0 \mid \psi_0] + \frac{1}{2}\Pr[1 \mid \psi_1]\right)_{|w'\rangle} &= \frac{1}{2} - \frac{1}{2\langle v|v\rangle}(\langle\psi_1|v\rangle\langle v|\psi_1\rangle - \langle\psi_0|v\rangle\langle v|\psi_0\rangle) \\
&> \frac{1}{2} + \frac{1}{2}(\langle\psi_1|v\rangle\langle v|\psi_1\rangle - \langle\psi_0|v\rangle\langle v|\psi_0\rangle) \tag{0.12} \\
&= \left(\frac{1}{2}\Pr[0 \mid \psi_0] + \frac{1}{2}\Pr[1 \mid \psi_1]\right)_{|u\rangle}
\end{aligned}$$

Hence, one can choose states in  $V$ , such that the distinguishing success probability is at least  $(\frac{1}{2}\Pr[0 \mid \psi_0] + \frac{1}{2}\Pr[1 \mid \psi_1])_{|u\rangle}$ .

Since for any normalized  $|u\rangle \in \mathcal{H}$ , in either cases, one can always choose normalized state  $|u'\rangle \in V = \text{span}\{|\psi_0\rangle, |\psi_1\rangle\}$  such that the distinguishing success probability is at least the same as  $|u\rangle$ , for maximizing the probability, one just need to consider  $\text{span}\{|\psi_0\rangle, |\psi_1\rangle\}$ .

- (ii) There is one trivial case we'll deal with first (to prevent exception). Suppose  $\langle\psi_0|\psi_1\rangle = 0$  (i.e. the two states are orthonormal), then choosing  $|u\rangle := |\psi_1\rangle$  (where if  $|v\rangle \in V$  and  $|w\rangle \in V^\perp$  has  $|u\rangle = |v\rangle + |w\rangle$ , then  $\langle w|v\rangle = 0$  and  $\langle v|v\rangle = \langle u|u\rangle$ ) has the above distinguishing success probability becomes the following:

$$\begin{aligned}
\left(\frac{1}{2}\Pr[0 \mid \psi_0] + \frac{1}{2}\Pr[1 \mid \psi_1]\right)_{|u\rangle} &= \frac{1}{2} + \frac{1}{2}(\langle \psi_1 | v \rangle \langle v | \psi_1 \rangle - \langle \psi_0 | v \rangle \langle v | \psi_0 \rangle) \\
&= \frac{1}{2} + \frac{1}{2}(1 - 0) = 1
\end{aligned} \tag{0.13}$$

Which, the probability can be maximized to 1, and it's definitely an optimal measurement.

Afterward, we'll assume  $\langle \psi_0 | \psi_1 \rangle \neq 0$ . Which, because  $|\psi_0\rangle, |\psi_1\rangle$  are normalized states which are linearly independent, so  $|\langle \psi_0 | \psi_1 \rangle|^2 < \langle \psi_0 | \psi_0 \rangle \langle \psi_1 | \psi_1 \rangle = 1$  using Cauchy Schwartz, and it's strict inequality because of linear independence. Hence, one has  $0 < |\langle \psi_0 | \psi_1 \rangle| < 1$ .

Given  $|u\rangle = x|\psi_0\rangle + y|\psi_1\rangle$ . WLOG, by multiplying suitable global phase, can assume  $x \geq 0$ . We'll also rewrite  $y = re^{it}$ , and  $\langle \psi_0 | \psi_1 \rangle = |\langle \psi_0 | \psi_1 \rangle|e^{is}$  for  $r \geq 0$  and  $t, s \in \mathbb{R}$ . Notice its norm is given as follow:

$$\begin{aligned}
\langle u | u \rangle &= x^2 \langle \psi_0 | \psi_0 \rangle + xy \langle \psi_0 | \psi_1 \rangle + y^* x \langle \psi_1 | \psi_0 \rangle + y^* y \langle \psi_1 | \psi_1 \rangle \\
&= x^2 + |y|^2 + 2x \operatorname{Re}(y \langle \psi_0 | \psi_1 \rangle) \\
&= x^2 + r^2 + 2x \operatorname{Re}(r |\langle \psi_0 | \psi_1 \rangle| e^{i(s+t)}) \\
&= x^2 + r^2 + 2xr |\langle \psi_0 | \psi_1 \rangle| \cos(s+t)
\end{aligned} \tag{0.14}$$

So, for  $|u\rangle$  to be normalized, one first need  $x^2 + r^2 + 2xr |\langle \psi_0 | \psi_1 \rangle| \cos(s+t) = 1$ .

On the other hand, the distinguishing success probability of  $|u\rangle$  is given as:

$$\begin{aligned}
\left(\frac{1}{2}\Pr[0 \mid \psi_0] + \frac{1}{2}\Pr[1 \mid \psi_1]\right)_{|u\rangle} &= \frac{1}{2} + \frac{1}{2}(\langle \psi_1 | u \rangle \langle u | \psi_1 \rangle - \langle \psi_0 | u \rangle \langle u | \psi_0 \rangle) \\
&= \frac{1}{2} + \frac{1}{2}(|y + x \langle \psi_1 | \psi_0 \rangle|^2 - |x + y \langle \psi_0 | \psi_1 \rangle|^2) \\
&= \frac{1}{2} + \frac{1}{2}(yy^* + xy \langle \psi_0 | \psi_1 \rangle + xy^* \langle \psi_1 | \psi_0 \rangle + x^2 |\langle \psi_0 | \psi_1 \rangle|^2) \\
&\quad - \frac{1}{2}(x^2 + xy^* \langle \psi_1 | \psi_0 \rangle + xy \langle \psi_0 | \psi_1 \rangle + yy^* |\langle \psi_0 | \psi_1 \rangle|^2) \\
&= \frac{1}{2} + \frac{1}{2}(1 - |\langle \psi_0 | \psi_1 \rangle|^2)(|y|^2 - x^2) \\
&= \frac{1}{2} + \frac{1}{2}(1 - |\langle \psi_0 | \psi_1 \rangle|^2)(r^2 - x^2)
\end{aligned} \tag{0.15}$$

With  $0 \leq |\langle \psi_0 | \psi_1 \rangle|^2 \leq \langle \psi_0 | \psi_0 \rangle \langle \psi_1 | \psi_1 \rangle = 1$  by Cauchy Schwartz (in fact,  $|\langle \psi_0 | \psi_1 \rangle|^2 < 1$  because  $|\psi_0\rangle, |\psi_1\rangle$  are linearly independent), the term  $(r^2 - x^2)$  has nonnegative coefficient. Hence, to maximize the probability, it suffices to maximize  $r^2 - x^2$ .

Now, we'll denote  $\theta := s + t$  (where  $t$  is the changing variable, but for simplicity purpose), and also  $a := |\langle \psi_0 | \psi_1 \rangle|$  to simplify the equation (where  $0 < a < 1$ ).

Then, the above question becomes: minimizing  $f(r, x, \theta) = r^2 - x^2$ , given the constraint  $g(r, x, \theta) = x^2 + r^2 + 2xra \cos(\theta) - 1 = 0$ . Hence, using Lagrange Multiplier, if  $(x, r, \theta)$  maximizes  $f$  while satisfying the constraint, there exists a unique  $\lambda \in \mathbb{R}$ , such that  $\nabla f(x, r, \theta) = \lambda \nabla g(x, r, \theta)$ . Which, we get the following equation:

$$(-2x, 2r, 0) = \nabla f = \lambda \nabla g = \lambda(2x + 2ra \cos(\theta), 2r + 2xa \cos(\theta), -2xra \sin(\theta)) \tag{0.16}$$

Since  $|u\rangle$  is assumed to be a nonzero state, one must have one of  $x, r = |y|$  be nonzero. In particular, this forces  $\lambda \neq 0$  (as if  $\lambda = 0$ ,  $\nabla f = (-2x, 2r, 0) = 0$ , enforcing  $x, r = 0$ , which is a contradiction).

Then, the third entry of the equality implies that  $-2\lambda x r a \sin(\theta) = 0$ , or  $x r \sin(\theta) = 0$ . There are three case to consider, either  $x = 0$ ,  $r = 0$ , or  $\sin(\theta) = 0$ :

- If  $x = 0$ ,  $r \neq 0$  is being enforced. Plug it into  $g$ , one has the following equality:

$$g(r, x, \theta) = x^2 + r^2 + 2xra \cos(\theta) - 1 = r^2 = 0 \quad (0.17)$$

Which implies that  $r = 0$ , but this is a contradiction to  $r \neq 0$ .

- If  $r = 0$ ,  $x \neq 0$  is being enforced. Plug it into  $g$ , one has the following equality:

$$g(r, x, \theta) = x^2 + r^2 + 2xra \cos(\theta) - 1 = x^2 = 0 \quad (0.18)$$

Which implies that  $x = 0$ , but this again contradicts with  $x \neq 0$ .

Hence, the only possibility is having  $\sin(\theta) = 0$  and  $x, r \neq 0$ , enforcing  $\cos(\theta) = \pm_1 1$ . Plug this result into the first two entries of the Lagrange Multiplier, we get:

$$-2x = \lambda(2x \pm_1 2ra), \quad 2r = \lambda(2r \pm_1 2xa) \quad (0.19)$$

where the  $\pm$  is by  $\cos(\theta)$  (which hasn't been determined yet). Then, the two equalities simplify to the following:

$$-x = \lambda x \pm_1 \lambda ar, \quad r = \lambda r \pm_1 \lambda ax \quad (0.20)$$

Which, the second equality states  $(1 - \lambda)r = \pm_1 \lambda ax$ . With  $\lambda, a, r, x \neq 0$ , this enforces  $1 - \lambda \neq 0$  (or  $\lambda \neq 1$ ). Hence, one gets  $r = \pm \frac{\lambda ax}{1 - \lambda}$ . Plugin to the first equality (which shows  $-(1 + \lambda)x = \pm_1 \lambda ar$ , or  $x = \mp_1 \frac{\lambda ar}{1 + \lambda}$ ), one has:

$$x = \mp_1 \frac{\lambda ar}{1 + \lambda} = -\frac{\lambda^2 a^2 x}{1 - \lambda^2} \quad (0.21)$$

Hence,  $(1 - \lambda^2)x = -\lambda^2 a^2 x$ , or  $(1 - \lambda^2(1 - a^2))x = 0$ . With  $x \neq 0$ , this enforces  $1 - \lambda^2(1 - a^2) = 0$ , or  $\lambda = \pm_2 \frac{1}{\sqrt{1 - a^2}}$ .

Here, plug the two potential  $\lambda$ 's back to the previous equation, we get:

$$\begin{cases} \left(1 \pm_2 \frac{1}{\sqrt{1 - a^2}}\right)x \pm_1 \left(\pm_2 \frac{a}{\sqrt{1 - a^2}}\right)r = 0 \\ \pm_1 \left(\pm_2 \frac{a}{\sqrt{1 - a^2}}\right)x + \left(\pm_2 \frac{1}{\sqrt{1 - a^2}} - 1\right)r = 0 \end{cases} \quad (0.22)$$

Because of the requirement  $x, r \neq 0$ , such linear system must have nontrivial solution. Which, for any  $x > 0$ , the solution is  $r = \mp_1 \left(\pm_2 \frac{\sqrt{1 - a^2}}{a}\right) \left(1 \pm_2 \frac{1}{\sqrt{1 - a^2}}\right)x$  (one can check this does satisfy the linear system).

Also, note that we have the restriction  $r \geq 0$  initially. With  $0 < a < 1$ , one has  $1 - \frac{1}{\sqrt{1 - a^2}} < 0$ . Hence, for  $r > 0$  to happen, the only possible pairs are  $(\pm_1, \pm_2) = (-, +), (-, -)$ . In particular, one needs  $\cos(\theta) = -1$  (so, in particular one can choose  $\theta = \pi$ ).

Finally, the two cases are given as follow:

- For  $\pm_2 = +$  (or  $\lambda = \frac{1}{\sqrt{1-a^2}}$ ), the solution is  $r = \frac{\sqrt{1-a^2}}{a} \left(1 + \frac{1}{\sqrt{1-a^2}}\right)x$ . Plugin to the equation  $g(x, r, \theta) = 0$ , one has:

$$\begin{aligned}
g(x, r, \theta) &= 0 \\
\Rightarrow x^2 + \frac{1-a^2}{a^2} \left(1 + \frac{1}{\sqrt{1-a^2}}\right)^2 x^2 - 2\sqrt{1-a^2} \left(1 + \frac{1}{\sqrt{1-a^2}}\right) x^2 &= 1 \\
\Rightarrow x^2 &= \frac{1}{2} \frac{a^2}{1-a^2} \frac{1}{1 + \sqrt{1-a^2}} \\
\Rightarrow r^2 - x^2 &= \left[ \frac{1-a^2}{a^2} \left(1 + \frac{1}{\sqrt{1-a^2}}\right)^2 - 1 \right] x^2 = \frac{1}{\sqrt{1-a^2}}
\end{aligned} \tag{0.23}$$

Hence, the distinguishing success probability for this case is:

$$\frac{1}{2} + \frac{1}{2}(1-a^2)(r^2 - x^2) = \frac{1}{2} + \frac{1}{2}\sqrt{1-a^2} \tag{0.24}$$

- for  $\pm_2 = -$  (or  $\lambda = -\frac{1}{\sqrt{1-a^2}}$ ), the solution is  $r = \frac{\sqrt{1-a^2}}{a} \left(\frac{1}{\sqrt{1-a^2}} - 1\right)x$ . Plugin to the equation  $g(x, r, \theta) = 0$ , one has:

$$\begin{aligned}
g(x, r, \theta) &= 0 \\
\Rightarrow x^2 + \frac{1-a^2}{a^2} \left(\frac{1}{\sqrt{1-a^2}} - 1\right)^2 x^2 - 2\sqrt{1-a^2} \left(\frac{1}{\sqrt{1-a^2}} - 1\right) x^2 &= 1 \\
\Rightarrow x^2 &= \frac{1}{2} \frac{1}{1 - \sqrt{1-a^2}} \frac{a^2}{1-a^2} \\
\Rightarrow r^2 - x^2 &= \left[ \frac{1-a^2}{a^2} \left(\frac{1}{\sqrt{1-a^2}} - 1\right)^2 - 1 \right] x^2 = -\frac{1}{\sqrt{1-a^2}}
\end{aligned} \tag{0.25}$$

Hence, the distinguishing success probability for this case is:

$$\frac{1}{2} + \frac{1}{2}(1-a^2)(r^2 - x^2) = \frac{1}{2} - \frac{1}{2}\sqrt{1-a^2} \tag{0.26}$$

Hence, for  $\pm_2 = +$ , one maximizes the the distinguishing success probability. Which, one has  $x = \frac{a}{\sqrt{2}\sqrt{1-a^2}\sqrt{1+\sqrt{1-a^2}}}$ , and  $r = \frac{\sqrt{1+\sqrt{1-a^2}}}{\sqrt{2}\sqrt{1-a^2}}$ .

Also, recall that we need  $\cos(\theta) = \cos(s+t) = -1$ , in particular we chose  $\theta = \pi$ . Hence, one has  $t = \pi - s$  (where  $s \in \mathbb{R}$  satisfies  $\langle \psi_0 | \psi_1 \rangle = |\langle \psi_0 | \psi_1 \rangle| e^{is} = ae^{is}$ ). So, the optimal measurement has state  $|u\rangle$  being the following (given that  $\langle \psi_0 | \psi_1 \rangle = ae^{is}$  for some  $0 < a < 1$  and  $s \in \mathbb{R}$ ):

$$\begin{aligned}
|u\rangle &= x|\psi_0\rangle + re^{it}|\psi_1\rangle \\
&= \frac{a}{\sqrt{2(1-a^2)(1+\sqrt{1-a^2})}} |\psi_0\rangle - \frac{\sqrt{1+\sqrt{1-a^2}}}{\sqrt{2(1-a^2)}} e^{-is} |\psi_1\rangle
\end{aligned} \tag{0.27}$$

And, the maximum distinguishing success probability is  $\frac{1}{2} + \frac{1}{2}\sqrt{1-a^2}$  shown by the Lagrange Multiplier.